

Theorem. (Erdős #12(ii)) There exists an infinite set $A \subset \mathbb{Z}^+$ with no distinct $a, b, c \in A$ with $b, c > a$, and $a|b+c$, yet for any $\varepsilon > 0$ and all sufficiently large N :

$$|A \cap \{1, \dots, N\}| \geq N^{1-\varepsilon}.$$

Consequently, there is no absolute constant $c > 0$ such that every such set A satisfies $|A \cap \{1, \dots, N\}| < N^{1-c}$ for infinitely many N .

Proof. The proof is similar to the previous one for Erdős #12(i), differing mainly in the use of a Behrend-style construction to produce a dense 3-AP-free set.

Let $c \in (0, 1)$, and let (P_k) be a sequence of pairwise coprime integers with $P_k \geq 3$ and $P_k \leq 4^{k+2}$ (for example, we can choose P_k to be the k -th odd prime). Next, define the running product $L_k = \prod_{i=0}^k P_i$. As before, we have $L_k \leq 4^{(k+2)^2}$. By the Chinese Remainder Theorem, there are integers $R_k < L_k$ satisfying $R_k \equiv 0 \pmod{P_k}$ and $R_k \equiv 1 \pmod{P_i}$ for all $i < k$.

Choose an integer $m \geq 2$ satisfying $m > (2m+1)^{1-c/2}$ and define the dimension $V_k = (k+10)^4$. We use a construction of Behrend to generate dense sets free of 3-term arithmetic progressions. Consider the grid of vectors $\{1, \dots, m\}^{V_k-1}$. Let $v = (v_0, \dots, v_{V_k-2})$ be such a vector. The squared norm $\|v\|_2^2$ takes integer values in the range $[V_k - 1, (V_k - 1)m^2]$. By the pigeonhole principle, there exists an integer radius K such that the number of vectors on this sphere satisfies $|\{v \in \{1, \dots, m\}^{V_k-1} \mid \|v\|_2^2 = K\}| \geq m^{V_k-1}/(V_k m^2 + 1)$.

Define S_k as the set of integers formed by evaluating these vectors in base $2m+1$, strictly offset by adding a massive leading digit $m(2m+1)^{V_k-1}$:

$$S_k = \left\{ m(2m+1)^{V_k-1} + \sum_{i=0}^{V_k-2} v_i(2m+1)^i \mid v \in \{1, \dots, m\}^{V_k-1}, \sum_{i=0}^{V_k-2} v_i^2 = K \right\}$$

Because the digits are bounded by m , adding two elements in base $2m+1$ involves no carrying. The restriction to the sphere of radius $\sqrt{K}$ ensures S_k contains no 3-APs. Furthermore, because the vectors use digits strictly between 1 and m , every element $x \in S_k$ satisfies $m(2m+1)^{V_k-1} \leq x < (m+1)(2m+1)^{V_k-1}$, ensuring the set is tightly clustered.

We define the blocks $A_k = \{L_k x + R_k \mid x \in S_k\}$, and set $A = \bigcup_{k \in \mathbb{N}} A_k$. Clearly, A is an infinite set. We verify A contains no distinct a, b, c such that $a \mid (b+c)$ with $b, c > a$. If a, b, c are in different blocks, a similar modular arithmetic argument as before using the CRT conditions yields a contradiction. If they are in the same block, the tightness of S_k guarantees $b+c < 3a$, forcing the divisibility multiplier to be exactly 2 (i.e., $2a = b+c$), which contradicts the fact that S_k is 3-AP-free. To ensure the blocks A_k do not overlap, we bound their elements. The maximum element in A_k is bounded above by:

$$\max(A_k) \leq L_k(m+1)(2m+1)^{V_k-1}$$

The minimum element in the next block A_{k+1} is bounded below by:

$$\min(A_{k+1}) \geq L_{k+1}m(2m+1)^{V_{k+1}-1}$$

Because the dimension $V_{k+1} = (k+11)^4$ is significantly larger than $V_k = (k+10)^4$, the base exponent strictly dominates, separating the blocks.

We now show that $|A \cap \{1, \dots, N\}| \geq N^{1-c}$ for sufficiently large N . The density drops the most in the empty gaps between blocks, with the absolute minimum occurring immediately before block A_{k+1} begins. At this point, $N = \min(A_{k+1}) - 1$, and our total element count is strictly greater than $|A_k|$. Thus, it suffices to prove:

$$|A_k| \geq (\max(A_{k+1}))^{1-c}$$

Substituting our parameter bounds into this inequality yields, and since m was chosen to satisfy $m > (2m+1)^{1-c/2}$, it remains to prove

$$\frac{((2m+1)^{1-c/2})^{V_k-1}}{V_k m^2 + 1} \geq (L_{k+1}(m+1)(2m+1)^{V_{k+1}-1})^{1-c}$$

Ignoring the polynomial denominator and constant multipliers, we compare the asymptotic growth of the exponents on the base $2m+1$:

- On the left hand side, the exponent is $(1-c/2)(V_k-1) = (1-c/2)k^4 + O(k^3)$.
- On the right hand side, the exponent is $(1-c)(V_{k+1}-1) = (1-c)(k+11)^4 = (1-c)k^4 + O(k^3)$.

The contribution of the running product $L_{k+1} \leq 4^{(k+3)^2}$ and the constant $(m+1)$ to the right hand side is only $O(k^2)$, which is safely absorbed into the $O(k^3)$ error term. Because $c > 0$, the k^4 coefficient on the left hand side $(1-c/2)$ is strictly greater than the k^4 coefficient on the right hand side $(1-c)$. Thus, for sufficiently large k , the left hand side asymptotically dominates, and the inequality holds. $\square$